

Collision in a circle

Y16 (2016) — English translation

On the air hockey table there are N identical small pucks, evenly arranged in a semicircle (see picture); the total mass of the washers is M . Another small washer, disc D , of mass m , slides in a direction perpendicular to the diameter of this semicircle and collides with the first of the stationary washers. By some miracle it happened that later it collided with all the other $N - 1$ washers in order, after which it continued to move in the direction opposite to the initial one. All collisions are absolutely elastic; friction can be neglected everywhere.

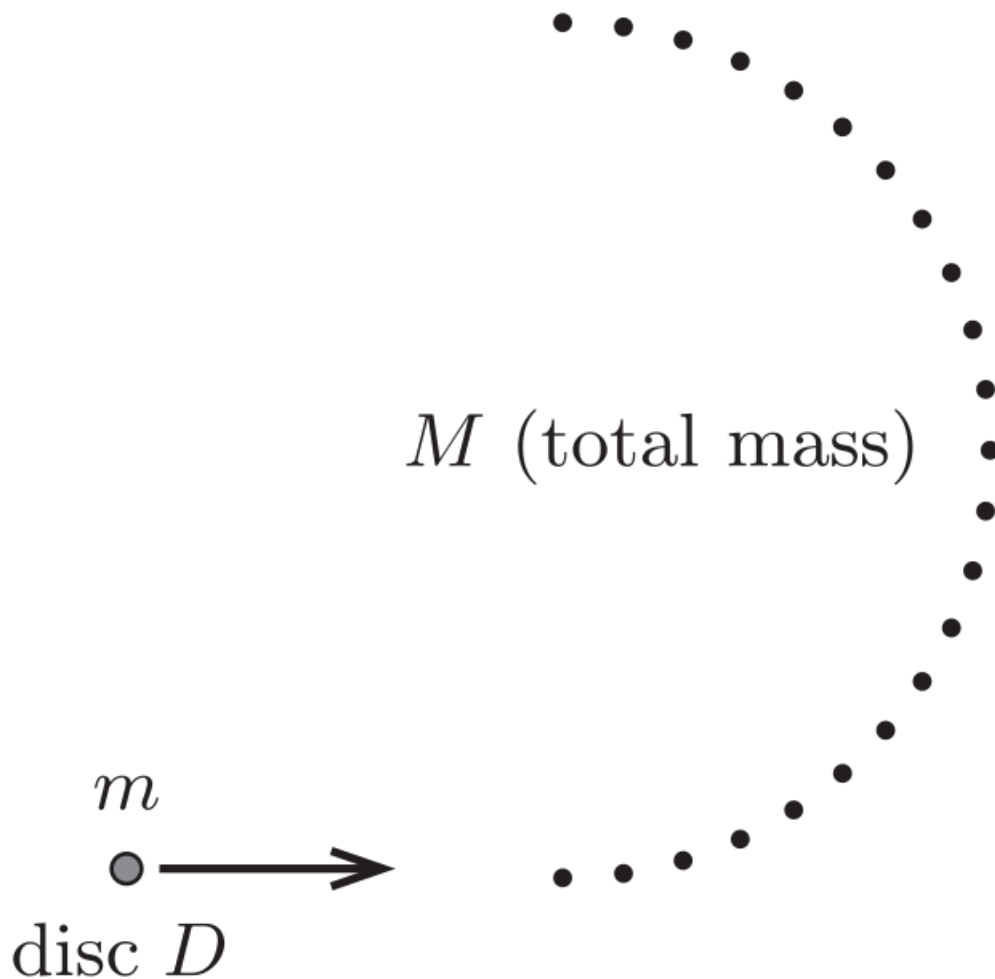


Fig. 1.

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| a | In the limiting case at $N \rightarrow \infty$, what is the minimum value of the ratio M/m at which the above miracle is possible? | 0 |
| b | If the mass ratio is equal to the critical value found in part a), what is the ratio of the final and initial speed D ? | 0 |